

National University of Computer and Emerging Sciences, Lahore Campus



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 Questions: 5

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Instruction/Notes:

- Don't ask any questions; in case of an ambiguity, use valid assumptions if needed.
- State the **units** of each value where necessary.

Q1 A 2.6KV, 1.3-MVA, 0.8-PF-lagging, 50-Hz, 8 pole, Y-connected synchronous generator, has a synchronous reactance of 1.2Ω and an armature resistance of 0.2Ω . At 50 Hz, its friction and windage losses are 28 kW, and its core losses are 22 kW. The field circuit has a dc voltage of 220 V, and the maximum I_f is 10 A. The resistance of the field circuit is adjustable over the range from 25 to 210 Ω . Total field resistance has been adjusted to 40 Ω and a delta connected load having per phase impedance of $18.186 + j10.5$ ohm is connected to the generator. OCC of the generator is given below. [20] CLO 04

Find out the following quantities.

- 1) voltage regulation
- 2) efficiency
- 3) output torque of prime mover

$$= \tau_{app} = \frac{P_{in}}{w_m} = \frac{18.186 + j10.5}{3}$$

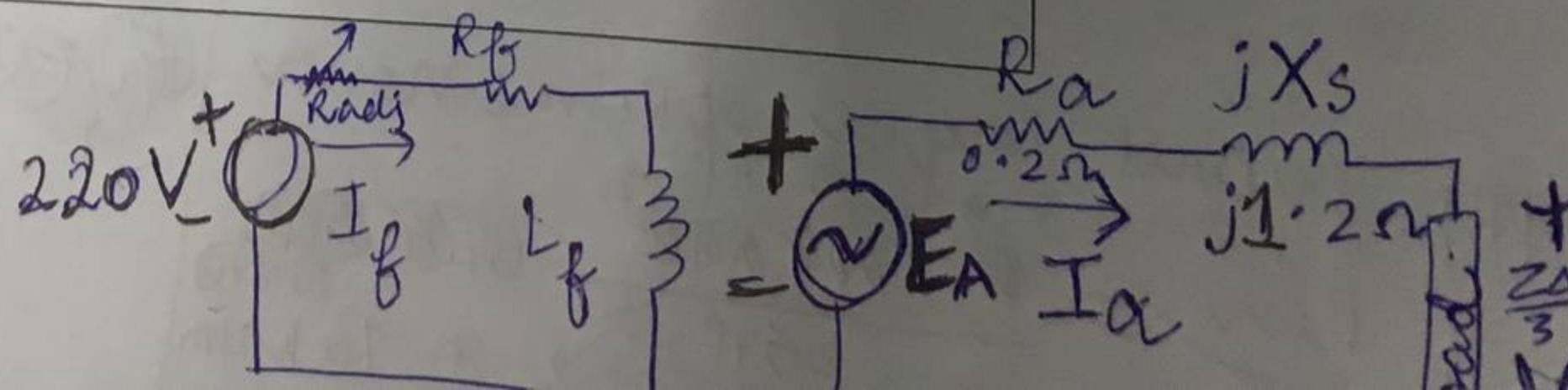
I_f (A)	2	4	6	8	10
$V_{T,NL}$ (V)	1200	2200	2600	2750	2850

OCC for above generator

$$Z_y = \frac{Z_{\Delta}}{3}$$

$$= \frac{18.186 + j10.5}{3}$$

$$Z_y = 6.062 + j3.5$$



Y-connected
 So $I_a = I_L$

$$V_L = 2.6 \text{ kV}$$

$$V_{\phi} = 1.501 \text{ kV}$$

$$S = 1.3 \text{ MVA}$$

$$\text{PF} = 0.8$$

$$f_c = 50 \text{ Hz}$$

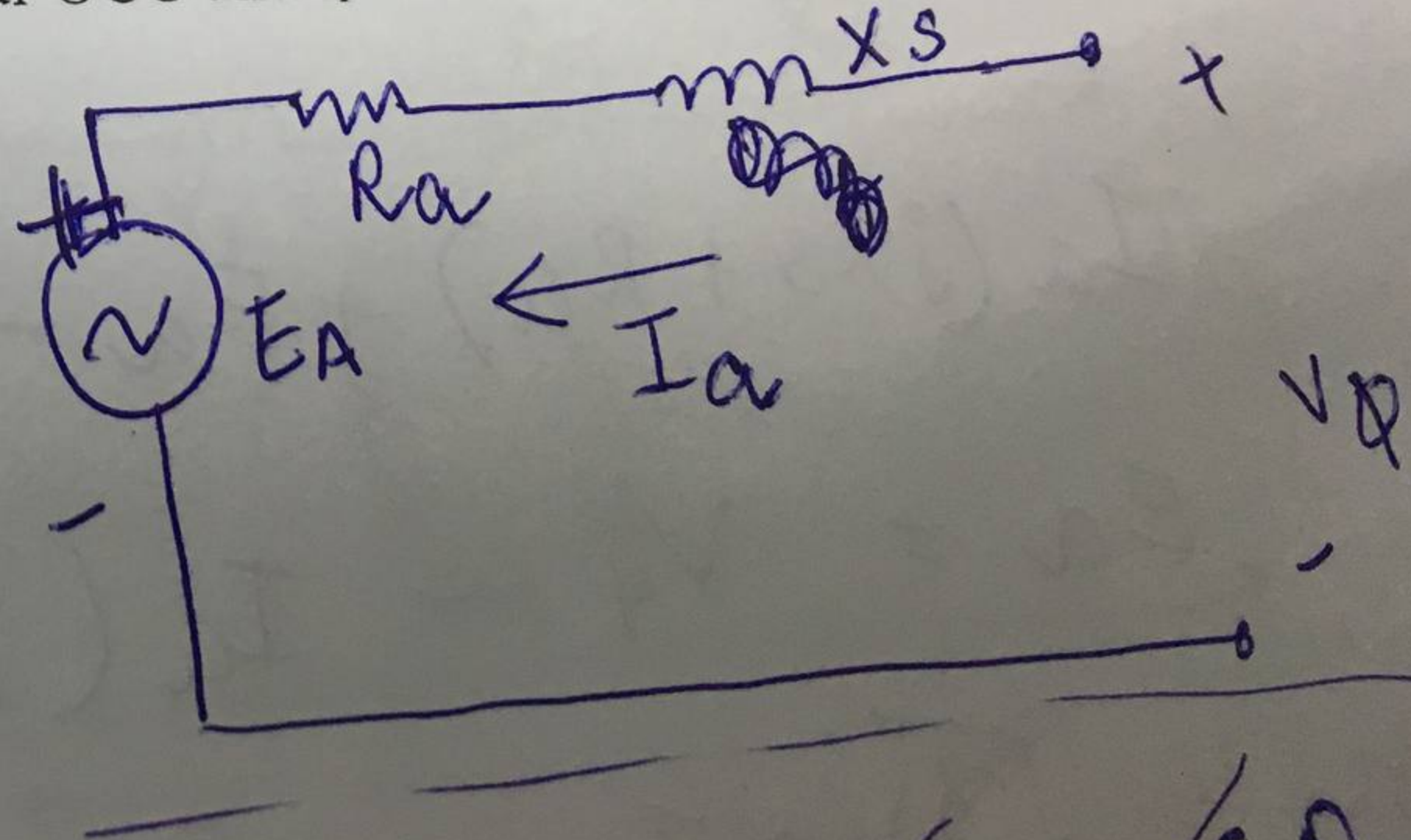
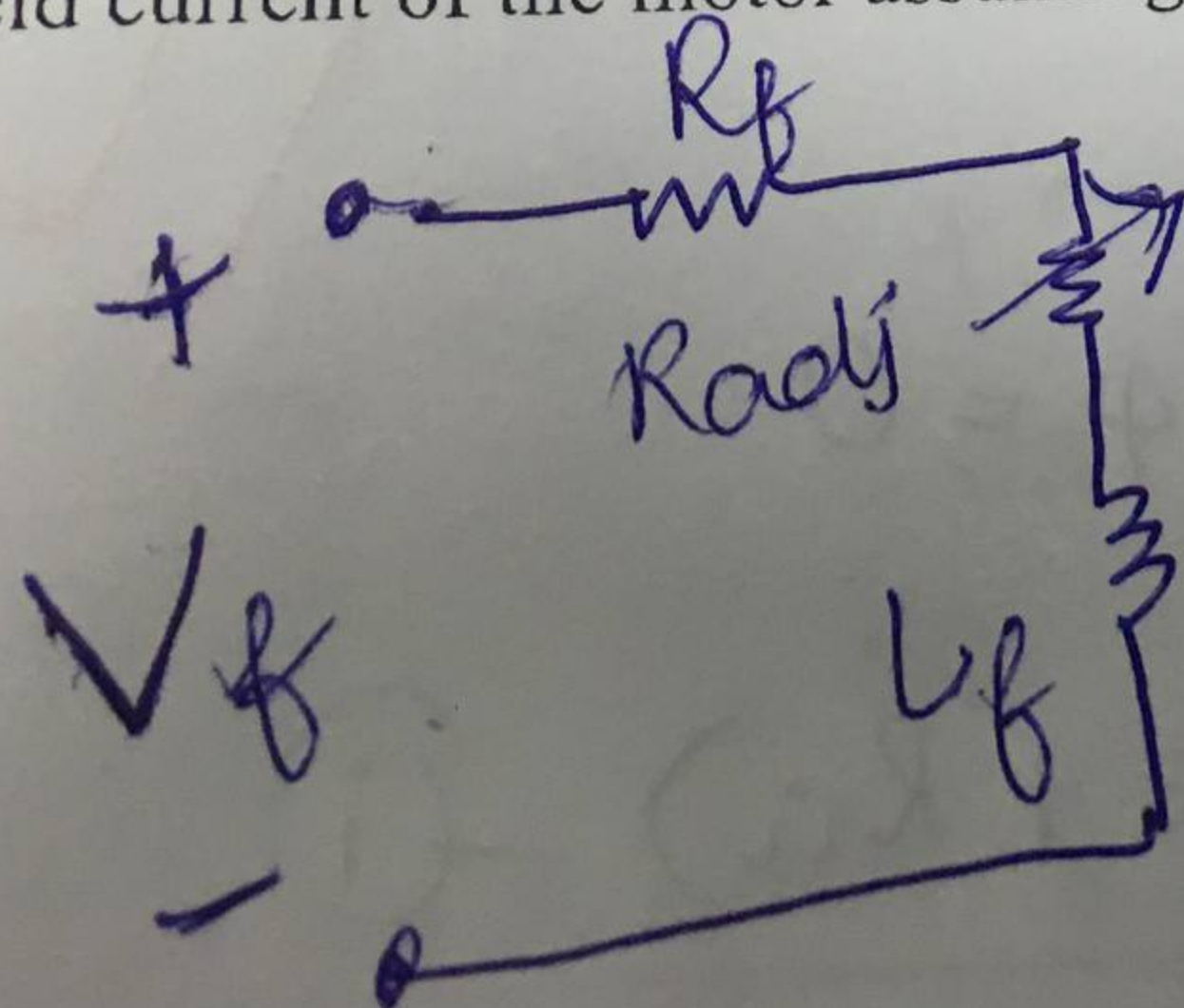
$$I_f (\text{max}) = 10 \text{ A}$$

$$R_{adj} = 25 - 210 \Omega$$

$$PF = \cos \left[\tan^{-1} \left(\frac{Q_T}{P_T} \right) \right]$$

Q2 An industrial load of 250KW is operating at a line voltage of 500 volts and a power factor of 0.7 lagging. In order to improve the power factor a synchronous motor, delivering an output power of 75KW, is connected in parallel to the load. The field current of the synchronous motor is adjusted so that the overall power factor becomes 0.9 lagging, while the total real power is now 325KW. Find out the following quantities ignoring losses of synchronous motor [20] CLO 05

- (a) Apparent power at which the synchronous motor is operating
 (b) Internally generated voltage of the Y connected motor if its synchronous reactance is 0.5Ω
 (c) Field current of the motor assuming linear OCC and $I_F = 5 \text{ A}$ at $E_A = 505.5 \text{ V}$



Q3 A 440-V 50-Hz two-pole Y-connected induction motor is rated at 100 kW. The [20]
equivalent circuit parameters are CLO

$$R_1 = 0.075 \Omega \quad R_2 = 0.065 \Omega \quad X_M = 7.2 \Omega$$

$$X_1 = 0.17 \Omega \quad X_2 = 0.17 \Omega$$

$$P_{F\&W} = 1.0 \text{ kW} \quad P_{\text{misc}} = 150 \text{ W} \quad P_{\text{CORE}} = 1.1 \text{ kW}$$

For a slip of 4% and at rated voltage and frequency, find out

SCRFM

06

- The rotor frequency
- The stator copper losses
- The air gap power

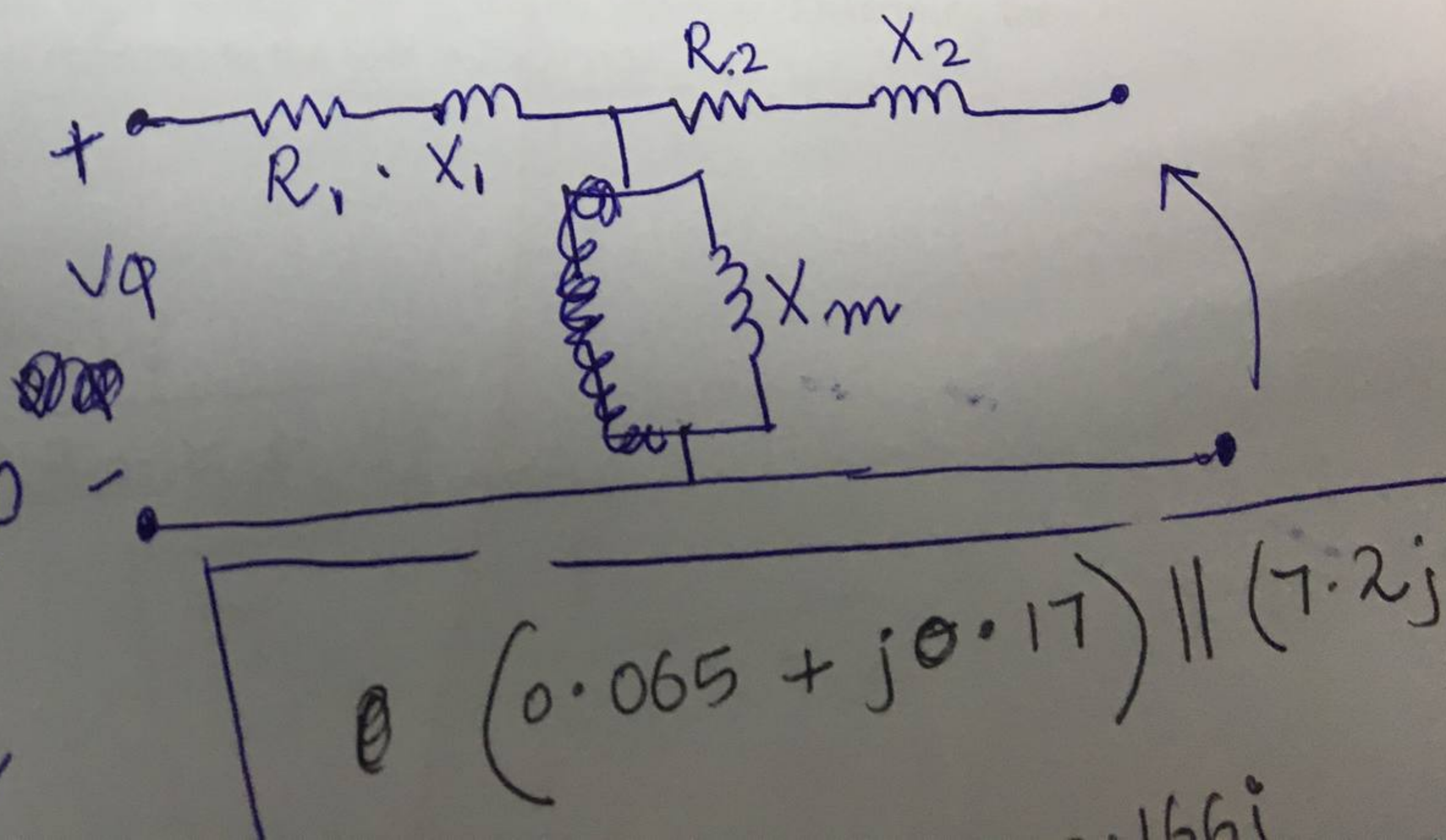
$$f_r = s \times f_e$$

$$= 4\%$$

$$= \frac{n_{\text{sync}} - n_m}{n_{\text{sync}}}$$

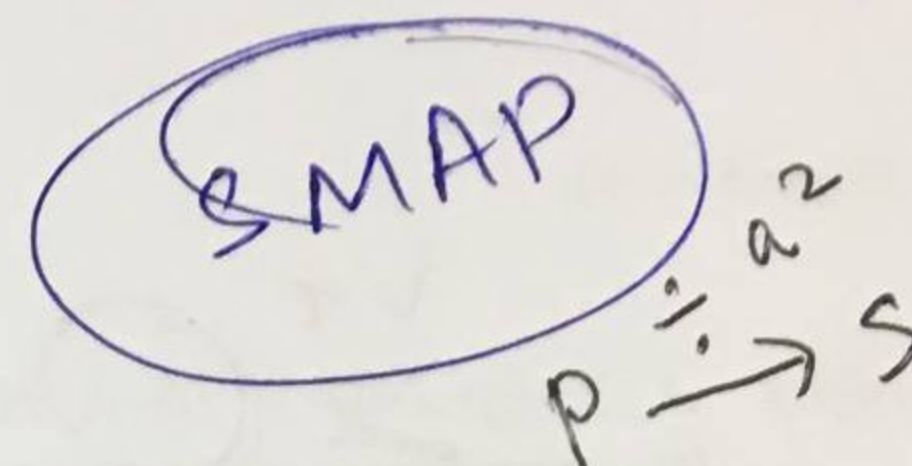
$$= \frac{f_e \times 120}{p} = \frac{50 \times 120}{2}$$

$$n_{\text{sync}} = 3000 \text{ rpm}$$



$$(0.065 + j0.17) \parallel (7.2j)$$

$$= 166i$$



S

Q4(a)

A 1000-VA 230/115-V, single phase transformer has been tested to determine its equivalent circuit. The results of the tests are shown below.

[10]
CLO

02

Open-Circuit Test**Short-Circuit Test**

$$V_{OC} = 230 \text{ V}$$

$$V_{SC} = 19.1 \text{ V}$$

$$I_{OC} = 0.45 \text{ A}$$

$$I_{SC} = 8.7 \text{ A}$$

$$P_{OC} = 30 \text{ W}$$

$$P_{SC} = 42.3 \text{ W}$$

Given data were taken from the high voltage side of the transformer. Determine the transformer's equivalent circuit referred to the low voltage side.

$$V_{OC} I_{OC} \cos \theta = P_{OC}$$

$$PF_{OC} = \frac{P_{OC}}{V_{OC} \times I_{OC}} = \frac{30 \text{ W}}{230 \times 0.45}$$

$$PF_{OC} = 0.289$$

$$PF_{SC} = \frac{P_{SC}}{V_{SC} \times I_{SC}} = \frac{42.3}{19.1 \times 8.7}$$

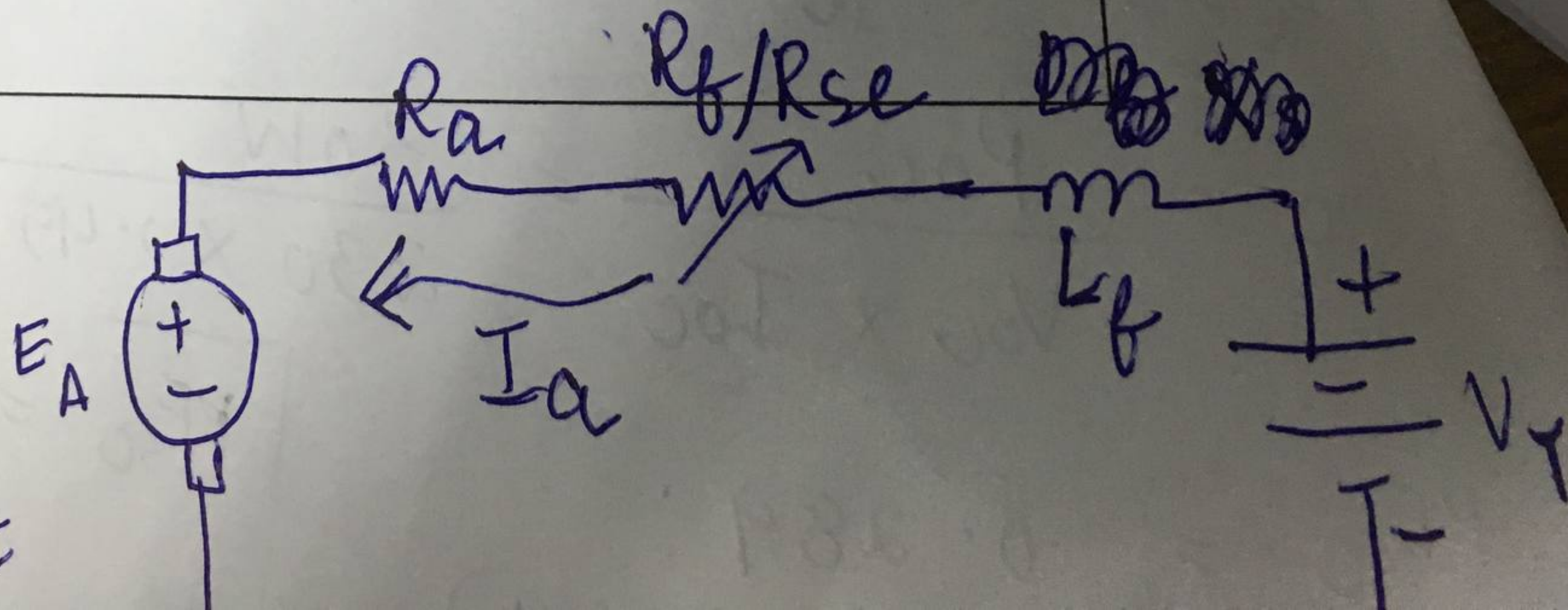
Q4(b) A 20-hp 240-V 76-A 900 r/min series motor has a field winding of 33 turns per pole. Its armature resistance is 0.09Ω , and its field resistance is 0.06Ω . The magnetization curve expressed in terms of magnetomotive force versus E_A at 900 r/min is given by the following table. Armature reaction is negligible in this machine. Compute the motor's output torque, speed, and output power at 133 percent of full-load armature current. (Neglect losses.)

E_A, V	95	150	188	212	229	243
$F, A \cdot \text{turns}$	500	1000	1500	2000	2500	3000

$I_a = 76 A$

3% of $I_a = 101.08 A$

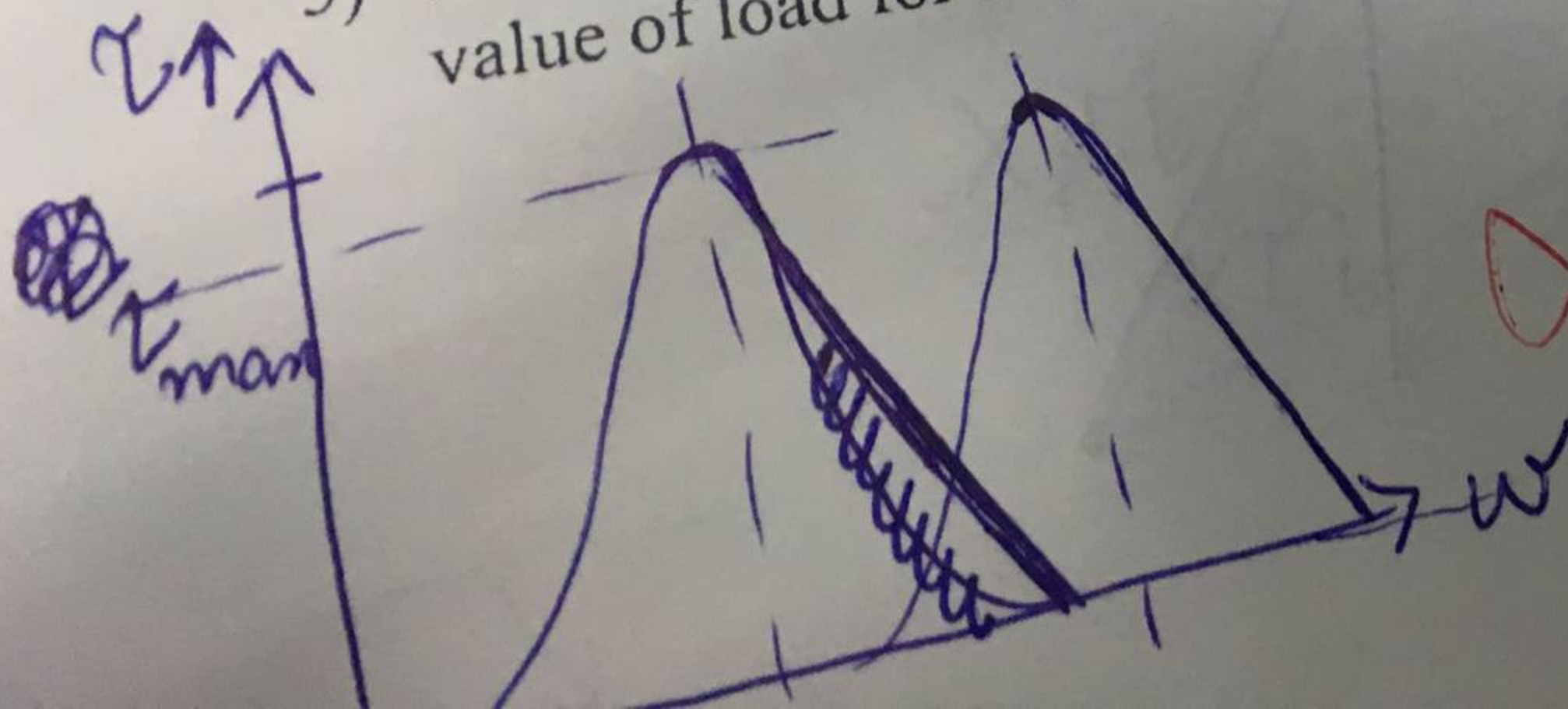
$= 76 \times 33 = 2508 A \cdot t$



Q5

Objective type: short a
1) Draw torque
resistance
2) A sym

- Q5** Objective type: short answers, labeled diagrams etc [Binary Marking] [20]
CLO
- 1) Draw torque speed curve of an induction motor for two different values of rotor resistance 06
 - 2) A synchronous generator has been connected in parallel with an infinite bus operating at 50 Hz, the generator has a no-load frequency of 52.5 Hz and a slope s_p of 1 MW/Hz. How much real power is being supplied by the generator 05
 - 3) Draw the torque speed curve of a synchronous motor
 - 4) Draw the torque speed curve of a compensated and uncompensated DC shunt motor on the same coordinate axis
 - 5) Draw a phasor diagram to show the impact of change in field current at a fixed value of load for a synchronous motor



R27R1

Formulae for synchronous machines

$$E_A^2 = (V_\phi + X_S I_A \sin \theta)^2 + (X_S I_A \cos \theta)^2$$

$$E_A = V_\phi + I_A(R_A + jX_S)$$

Induction Motors Power Calculations

$$P_{in} = \sqrt{3} V_L I_L \cos \theta = 3 V_{ph} I_{ph} \cos \theta$$

$$P_{SCL} = 3 I_1^2 R_1$$

$$P_{AG} = P_{in} - (P_{SCL} + P_{core})$$

$$P_{RCL} = 3 I_2^2 R_2 = s \times P_{AG}$$

$$P_{conv} = P_{AG} - P_{RCL}$$

$$P_{out} = P_{conv} - (P_{f+w} + P_{stray})$$

$$\tau_{ind} = \frac{P_{conv}}{\omega_m}$$

$(1-s) \times P_{AG}$